

Atharva Atul Rege

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EDUCATION

National Institute of Technology Karnataka, Surathkal <i>Bachelor of Technology in Computer Science and Engineering – 8.89 CGPA</i>	Mangaluru, India Aug 2023 – Present
Private International English School <i>CBSE Class 12 – 97.6/100</i>	Abu Dhabi, UAE Aug 2022 – Mar 2023

EXPERIENCE

I3D Lab, Indian Institute of Science (IISc) <i>Research Intern</i>	May 2025 – July 2025 Bengaluru, India
- Developed a two-stage diffusion-based image generative pipeline that integrates material attribute editing and semantic object placement for image augmentation, under the guidance of Prof. Pradipta Biswas. - Implemented a Material Property Editing module leveraging CLIP embeddings space interpolation to modify foreground object properties such as color, transparency, metallic nature, albedo, glow and surface texture. - Designed a Background Context-driven Object Placement framework predicting optimal 3D position, orientation, and scale for objects within a given background scene, followed by seamless inpainting.	

PUBLICATIONS

- **Improving MR Interaction through GenAI** | *ACM IUI 2026 (Under Review)*
Yashaswi Sinha, **Atharva Atul Rege**, Yash Kumar Sahu, Rubini M, Himanshu Vishwakarma, Pradipta Biswas

PROJECTS

The Brush of Spells : Text-Guided Image Inpainting <i>Python, PyTorch</i>	Nov 2024 – Apr 2025
- Implemented <i>MMFL: Multimodal Fusion Learning for Text-Guided Image Inpainting</i> using Generative Adversarial Networks (GANs), leveraging a multimodal architecture to inpaint images based on textual prompts. - Enhanced the base implementation by adding a CLIP-based loss and Wasserstein GAN (WGAN)-based loss to improve semantic alignment. - Trained the model on the CUB-200-2011 image-text dataset for better inpainting accuracy for images of birds.	
Vision Kinect <i>Python, OpenCV, TensorFlow, Keras</i>	Feb 2024 – May 2024
- Implemented a hand gesture recognition application using YOLO object detection algorithm and mapped each hand gesture to a specific in-game function or control in an interactive Tetris game. - Enabled fully touchless gameplay using gesture-based controls, demonstrating applications in interactive gaming and Human-Computer Interaction (HCI).	
FinBot-AI <i>Python, LangChain, Streamlit</i>	Sep 2024 – Nov 2024
- Developed a Generative AI-powered chatbot for finance-related queries, leveraging OpenAI's GPT-4, GPT-4-Turbo, and GPT-4o APIs to deliver accurate, context-aware responses. - Utilized the LangChain framework for efficient text generation and processing by developing a Retrieval-Augmented Generation (RAG)-based application with an interactive web interface.	

TECHNICAL SKILLS

Languages: Python, C, C++, Java, HTML, CSS
Libraries: PyTorch, TensorFlow, OpenCV, Numpy, Pandas, FastAPI, LangChain, scikit-learn
Technologies: MySQL, Docker, Git, Linux

EXTRACURRICULAR ACTIVITIES

IEEE - NITK Student Chapter <i>Executive Member</i>	Jan 2024 – Present Mangaluru, India
- Contributed to the successful organization of workshops, interactive sessions, and competitions, attracting nationwide participation. - Mentored a student-led project titled QuantumQuill: A Transformer-based Sci-Fi Storyteller focusing on fine-tuning Large Language Models (LLMs) for the generation of coherent and creative science fiction stories.	